RIG-SMOOTHNESS OVER ONE IDEAL OF DEFINITION IMPLIES ALL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the counted statement is the author lemma environment labelled lemma-rig-smooth-continuous; the proof environment immediately following it is its own proof

Editorial note. The label lemma-rig-smooth-continuous names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- ideals of definition of a Noetherian adic ring are pairwise commensurable: for any two ideals of definition I and I' there is c >= 0 with I^c subset I' and (I')^c subset I
- the definition of rig-smooth homomorphisms of WAdm^Noeth via the completed naive cotangent complex (definition-rig-smooth-homomorphism)
- the completed naive cotangent complex NL^wedge_{B/A} is independent of the chosen ideal of definition of A (remark-NL-well-defined-topological of the same chapter)

Source reference. Original reference 0GAI: sources/stacks-restricted/pinned-original-source.tex, source lines 831–831. Prerequisite pointer invoked at proof line(s) 4233; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0ANA: sources/stacks-chow/tags.txt, source lines 13433–13433. Prerequisite pointer invoked at proof line(s) 4216; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 04U5: sources/stacks-restricted/pinned-original-source.tex, source lines 2439–2439. Prerequisite pointer invoked at proof line(s) 4220; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GBX: sources/stacks-restricted/pinned-original-source.tex, source lines 2487–2487. Prerequisite pointer invoked at proof line(s) 4237; it names a standard fact of the same chapter that the counted argument

Date: stacks-project:restricted.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $A \to B$ be a morphism in $WAdm^{Noeth}$ (Formal Spaces, Section 0ANA). The following are equivalent:*

(a) *$A \to B$ satisfies the equivalent conditions of Lemma 04U5 and there exists an ideal of definition $I \subset B$ such that B is rig-smooth over (A, I), and*
(b) *$A \to B$ satisfies the equivalent conditions of Lemma 04U5 and for all ideals of definition $I \subset A$ the algebra B is rig-smooth over (A, I).*

Proof. Let I and I' be ideals of definitions of A. Then there exists an integer $c \geq 0$ such that $I^c \subset I'$ and $(I')^c \subset I$. Hence B is rig-smooth over (A, I) if and only if B is rig-smooth over (A, I'). This follows from Definition 0GAI, the inclusions $I^c \subset I'$ and $(I')^c \subset I$, and the fact that the naive cotangent complex $NL^\wedge_{B/A}$ is independent of the choice of ideal of definition of A by Remark 0GBX. □